

Perceived Realism in 360° Video-Based vs. 3D Modeled Virtual Reality Environment

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ABSTRACT

Virtual reality impacts users differently depending on the visualization environments. This study compares the impact of two visualization styles in virtual environments on perceived realism and perception. An experiment was conducted using a high-fidelity 360° video-based environment and a modeled three-dimensional (3D) environment. The results show that the 360° video-based environment produced a higher perceived realism rating and a greater sense of reality. However, no impact was observed on users' perception of the depicted environments. Users' perceptions were positive, indicating an engaging experience with a great effect on opinions regarding the stigma of Geelong's post-industrial manufacturing status.

Keywords: Virtual reality, fidelity, presence, realism, user perception, three-dimensional, 3D modeled.

Index Terms: Human-centered computing, Human-computer interaction, Virtual reality.

1 INTRODUCTION

Virtual reality (VR) provides an artificial three-dimensional (3D) environment, allowing users to interact and often simulate real-life scenarios [5]. The user experience in VR relating to the sense of presence or the feeling of 'being there' is essential to differentiate between VR systems [6]. Investigating the factors affecting presence in VR is critical to developing engaging virtual environments in human-computer interaction. Studies compared the effectiveness of immersive and non-immersive virtual environments, with some researchers having real environments as the baseline for the between-group studies. However, presence effectiveness in different immersive virtual environment settings has not been thoroughly assessed.

User perception refers to the cognitive dimensions of understanding other opinions extending beyond the ability to see, hear, or become aware of something through the senses [1]. This research highlights the significance of understanding the relationship between perceived realism and user perception. This paper elucidates how perceived realism facilitates content comprehension, which is crucial to the development of engaging and effective VR applications [3]. Therefore, it compares the impact of 3D modeling and 360° video-based virtual environments on perceived realism and users' perception.

The case study focuses on the post-industrial status and the stigma surrounding the manufacturing industry as dirty and noisy. The manufacturing facilities in the City of Geelong were assumed to be traditional due to the low access and exposure to the public. Furthermore, this study aims to quantify perceived realism and users' perception of the local manufacturing industry.

2 RESEARCH DESIGN

Both virtual environments had 360° vision and utilized the entire field of view of 110° provided by the HTC Vive Pro Eye headset. The 360° video-based environment was created by stitching images of the manufacturing facility (Figure 1, left picture). The 3D-modeled virtual environment was a virtually curated model (Figure 1, right photo).



Figure 1: Sample of the virtual environments used in this study.

This study utilized mixed research methods design using a between-group design to compare the impact of 3D modeled and 360° virtual environments on perception. Researchers recruited participants who are 18 and older students and staff with varying experience or knowledge about manufacturing in Geelong and Virtual Reality. All participants were randomly assigned to a group. The total duration of each experiment was around 40 minutes, which only included 5 minutes of immersion in virtual reality. The participants were required to stand during the virtual immersion experiment. Both participant groups had no control over depth, traversing, or exploration with fixed positioning to reduce distraction, maintain a consistent experience for all participants, and ensure the repeatability of this experiment.

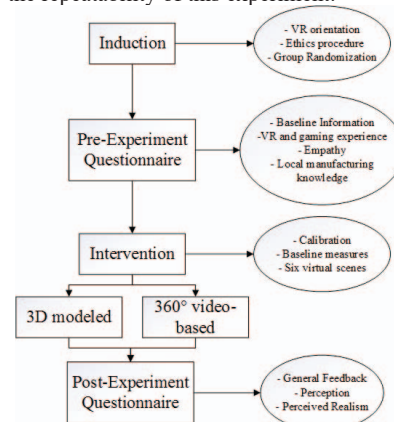


Figure 2: Experimental Protocol.

Participants completed a post-experimental survey after the virtual experiment intervention (Table 1) with seven-point Likert scale questions covering perceived realism and perception. This study used an established questionnaire (RQ) and added general questions common within the field of human-computer interaction [3].

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Table 1: Post-experiment survey.

Measures	Statements
Perceived realism	Realism Rating: In your opinion, how realistic was this environment? PR 1: To what extent did you feel that you were really inside the virtual manufacturing facility? PR 2: To what extent did you feel that you were surrounded by the virtual manufacturing equipment? PR 3: To what extent did you feel that you really visited the virtual manufacturing facility? PR 4: To what extent did you feel that the manufacturing facility seemed like the real world? PR 5: To what extent did you feel that could reach out and touch the virtual equipment?
Perception	P1: To what extent did you feel a change of your opinion about Manufacturing in Geelong? P2: To what extent did you feel that this experience impacts your point of view about manufacturing in Geelong? P3: How did this experience impact your point of view about manufacturing in Geelong?

3 RESULTS

Continuous data are presented as mean [standard deviation]. Statistical tests used the Mann-Whitney U test to compare means and standard deviations. Table 2 below presents perceived realism results comparing 3D modeled and 360° captured VR environment groups. All perceived realism measures use a Likert scale between one and seven. One represents a lack of realism, four is a neutral feeling of realism, and seven is a high sense of realism. The Realism Rating was a separate seven-point Likert scale item answered before the five realism items, weighing statistically equally. Realism total is calculated by adding the five items of realism. The Realism questionnaire showed high reliability with Cronbach's alpha $\alpha = 0.846$ and intraclass correlation coefficient = 0.524 ranging between 0.446 and 0.621.

These results show that the Realism Rating with a mean of (5.88[1.336]) and Realism-4 with a mean of (5.42[1.528]) in 360° video-based VR environment was significantly higher than the 3D modeled group Realism Rating of (5.07[1.542]) and Realism-4 of (4.67[1.593]). On the contrary, Realism subscale 5 had higher scores in the 3D modeled VR environment (5.19[1.642]) than in the 360° video-based environment (4.73[1.373]).

Like the perceived realism measures, the user perception questions were measured using a Likert seven-point scale. The three perception items had a high Cronbach's alpha $\alpha = 0.81$ and intraclass correlation coefficient = 0.588 (0.532 – 0.713). There was no significant difference in the mean perception scores between the two groups, as shown below in Table 2.

Table 2: Questionnaire results.

Measures	360° video-based	3D modeled	Sig.	r
Realism Rating	5.88 (1.336)	5.07 (1.542)	0.01*	0.3
Realism Total	30.27 (6.93)	30.74 (5.85)	0.47	0.01
PR-1	5.5 (1.393)	5.56 (1.34)	0.49	0.01
PR-2	5.23 (1.451)	5.67 (1.301)	0.15	0.14
PR-3	5.38 (1.416)	5.56 (1.086)	0.47	0.01
PR-4	5.42 (1.528)	4.67 (1.593)	0.03*	0.26
PR-5	4.73 (1.373)	5.19 (1.642)	0.04*	0.23
P-1	4 (1.697)	4.11 (1.717)	0.4	0.03
P-2	4.85 (1.69)	4.63 (1.597)	0.28	0.08
P-3	5.46 (1.067)	5.11 (1.502)	0.26	0.1

4 DISCUSSION

Perceived Realism questionnaires showed statistical differences between the virtual environments in the Realism Rating, PR - 4 and PR - 5 scores. Realism Rating and PR - 4 were higher in the 360° video-based VR environment group than in the 3D-modeled environment group. These results align with the conclusion of Mizuho et al. [2]. PR 5 was scored higher by 3D-modeled participants because this virtual environment represented better spatial awareness and depth cues.

These results highlight the impact of depth on perceived realism, which positively affects the understanding of representation if present and agrees with past studies [3]. These findings suggest that the depth impact on perception in 3D-modeled environments was more substantial than the effectiveness of details displayed in 360° video-based environments. These findings indicate that the objective of the VR application is a contributing factor in recommending a VR environment style. These findings reflect those of Pedram et al. [4]. Concerning user perception results, all scores were higher than 4 in both groups, showing the effectiveness of the VR technological impact on perception. In addition, the Perception-3 scores were 5.46 and 5.11 for the virtual environments, which resembles the positive impression the experience had on the participants' perception. The chi-square Spearman's correlation study found that user perception measures P2 and P3 positively correlate with perceived realism measures with coefficients ranging between 0.29 and 0.76. These coefficients emphasize the impact of perceived realism on shifts in user perception.

Further experimental investigations are needed to analyze the impact of varying levels of display fidelity in 360° digitized environments on perception and perceived realism to improve the theoretical understanding and methodological application of perceived realism. Comparing two 360° digitized VR environments will eliminate the effect of depth cue, focusing on perceived realism. The produced VR application will significantly expand the social reach of local manufacturers and stakeholders. Its primary aim is to aid in the re-branding and promoting the current manufacturing companies and technologies.

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